

Project: Patient Appointment & Triage Assistant

Objective

The objective of this project is to build a digital assistant that improves the efficiency and effectiveness of patient appointment scheduling and triage processes within healthcare organizations.

The system should ensure that patients are directed to the appropriate care pathway based on the urgency and nature of their symptoms, while optimizing the allocation of medical resources such as doctors, specialists, and appointment slots.

Additionally, the solution should enhance patient experience by reducing waiting times, improving access to care, and providing clear guidance throughout the appointment scheduling process, while strictly adhering to data privacy and healthcare regulations.

Description

Healthcare providers often face significant challenges in managing patient demand, particularly in distinguishing between urgent and non-urgent cases. Traditional scheduling systems typically operate on a first-come, first-served basis, which can lead to inefficiencies and delays in critical care.

This system aims to introduce an intelligent triage and scheduling assistant that collects patient information, including symptoms and urgency indicators, and analyzes this data to determine priority levels. Based on this assessment, the system will recommend appropriate next steps, such as booking an appointment with a general practitioner, referring to a specialist, or escalating to urgent care.

The scheduling component should provide visibility into available time slots, medical staff availability, and resource constraints, ensuring optimal allocation. Healthcare staff will have access to a dashboard where they can review, adjust, and manage appointments and triage decisions.

Given the sensitive nature of healthcare data, the system must ensure strict compliance with privacy regulations such as GDPR, including secure data handling, access control, and auditability.

The overall goal is to create a smarter, patient-centered system that improves care delivery while reducing administrative burden on healthcare professionals.

Requirements

- Allow patients to request appointments online
- Collect and analyze patient symptoms

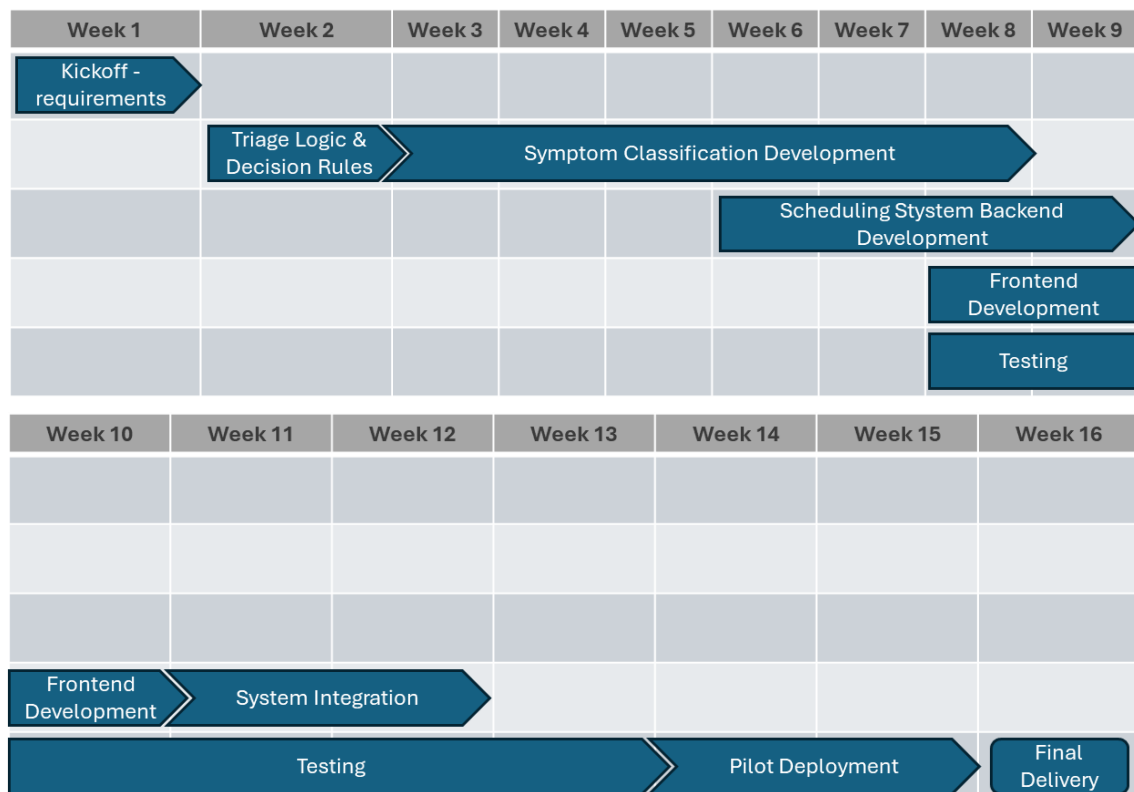
- Automatically prioritize cases based on urgency
- Suggest appropriate departments or specialists
- Provide a scheduling interface for staff
- Ensure data privacy and compliance (e.g., GDPR)

Deadlines

- MVP Delivery: 6 weeks
- Pilot Launch: 3 weeks

Milestones

1. Requirements gathering and compliance validation (Week 1)
2. Triage logic and decision rules definition (Week 2)
3. Symptom classification development (Week 3-8)
4. Scheduling system backend development (Week 6-9)
5. Frontend interface development (Week 8-10)
6. System integration (Week 11-12)
7. Testing and validation (Week 8-13)
8. Pilot deployment (Week 14-15)
9. Final delivery (Week 16)



Dependencies

- Triage depends on symptom classification logic
- Scheduling depends on resource availability data
- System must comply with data privacy regulations

Team Profiles

Core Roles

- Product Owner - defines vision, priorities, and patient experience goals
- Business Analyst - gathers requirements and maps clinical workflows
- Healthcare Specialist - defines triage rules and validates medical logic

Technical Roles

- Backend Developer - builds APIs, scheduling, and triage logic
- Frontend Developer - develops patient interface and staff dashboard
- Data Scientist / ML Engineer - develops symptom classification and prioritization logic

Compliance & Security

- Compliance / Data Privacy Specialist - ensures GDPR and healthcare regulation compliance
- Security Engineer - handles data protection and access control

Support Roles

- UX/UI Designer - designs patient journey and usability
- DevOps Engineer - manages infrastructure and deployments